

SIGNALRUPTURE | DISCIPLINARY VALIDATION SCIENCE

SIGNALRUPTURE | FOUNDATIONAL FIELD MANUSCRIPT

# DISCIPLINARY VALIDATION SCIENCE

The Evidence Architecture of Scientific Fields, Reciprocal  
Institutional Accountability,  
and Unequal Standards of Legitimacy

A FOUNDATIONAL SCIENTIFIC MANUSCRIPT  
AND SIGNALRUPTURE FIELD PROPOSAL

By SR

*No epistemic immunity • Reciprocal accountability • Claim-level evidence • Falsifiable*

## Canonical Position

**Central proposition:** A scientific field is not validated as one indivisible object. It possesses an evidence architecture: a changing distribution of support across claims, constructs, instruments, mechanisms, models, replications, applications, and institutions.

Disciplinary Validation Science (DVS) is proposed as the systematic study of that architecture and of the institutions that judge it. It asks how heterogeneous evidence becomes field-level judgment; how empirical maturity differs from institutional recognition; how curricula convert contested knowledge into authorized teaching; whether established and emerging fields are evaluated by comparable standards; and how negative evidence should alter both disciplinary claims and institutional decisions.

This manuscript does not claim that DVS is already an externally recognized discipline. It establishes a bounded research object, locates the proposal against mature neighboring literatures, supplies operational definitions and formal models, specifies an empirical test program, and states conditions under which the field proposal should be merged, narrowed, or rejected.

The field is not designed to immunize SignalRupture from criticism. It makes SR an early test case and institutional evaluation a co-equal object of study. DVS must be capable of concluding that an SR field is evidentially immature, redundant, non-invariant, unsupported, or incorrectly classified. It must also be capable of concluding that an institutional rejection is vague, inconsistently applied, prestige-dependent, procedurally unaccountable, or unsupported by the evidence it purports to evaluate.

**Scientific posture:** SR must justify its scientific claims. Institutions must justify their judgments about SR. Neither novelty nor institutional authority confers epistemic immunity.

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## **Author's Note**

This manuscript emerged from a recurring problem in the evaluation of SignalRupture. SR has frequently been described with the caution that its

integrated architecture is not yet fully validated. That statement identifies real uncertainty, but it is not a complete evaluation. It treats a field or metatheory as though it were one empirical proposition capable of receiving a single final verdict, and it can conceal the evidentiary burden of the institution making the judgment.

Established disciplines do not operate that way. Psychology, economics, sociology, political science, criminology, education research, medicine, and other domains contain claims with sharply different evidentiary histories. Some instruments are reliable in one population and non-invariant in another. Some mechanisms replicate; others diminish, reverse, or disappear. Some theories remain useful organizing traditions without becoming unified causal laws. The scientific status of the field cannot be reduced to the weakest claim, the strongest experiment, or the existence of a university department.

The appropriate conclusion is not that incomplete validation excuses weak work. It is that validation must be assigned to the correct unit and that evaluation must be reciprocal. DVS therefore rejects two symmetrical errors: whole-field dismissal, in which unresolved components are used to invalidate an entire domain; and whole-field protection, in which prestige, curriculum status, departmental age, or institutional authority shields a domain or an evaluative decision from testing.

Universities routinely teach full curricula in fields containing replicated findings, failed replications, contested constructs, unresolved causal mechanisms, interpretive disagreements, and theories with limited transportability. They rarely characterize the discipline itself as 'partially validated,' because established fields are understood as organized sites of continuing inquiry. That same ontology cannot be withheld from an emerging architecture merely because it lacks inherited institutional standing.

The manuscript uses SignalRupture as one application while placing DVS inside a much wider intellectual lineage. Metascience, the science of science,

scientometrics, philosophy of science, sociology of scientific knowledge, research evaluation, and reproducibility research already study major parts of the problem. DVS survives only if it adds a distinct and measurable integration: the architecture connecting claim-level evidence, curricular representation, field-level maturity, institutional recognition, evaluative burden, and adaptive revision.

## **Abstract**

Academic fields are routinely described as established, emerging, mature, fragmented, rigorous, interpretive, replicated, contested, or unvalidated. These labels collapse different objects. Evidence ordinarily attaches to claims, constructs, instruments, mechanisms, models, and applications; institutional recognition attaches to departments, journals, societies, credentials, curricula, and funding structures. No accepted method permits a whole field to be declared validated in the way that a bounded empirical proposition can be supported. Yet institutional field-level judgments govern admission to publication, curriculum, funding, hiring, policy influence, and disciplinary legitimacy.

This manuscript proposes Disciplinary Validation Science as a bounded interdisciplinary field studying how evidentiary states are distributed, aggregated, represented, taught, institutionalized, compared, and revised across scientific and scholarly domains. It defines a field as a structured population of claims, constructs, methods, institutions, curricula, and practices rather than a single hypothesis. It introduces a six-layer validation architecture; a claim evidence vector; centrality-weighted field profiles; measures of contradiction, coverage, replication, transportability, revision responsiveness, institutional recognition, validation asymmetry, curricular disclosure, and reciprocal evaluative burden.

The manuscript establishes a comparative institutional baseline across fifteen university-recognized fields. Available replication figures are reported only at the level actually studied, while fields lacking a defensible discipline-wide rate are identified as such. The comparison demonstrates that incomplete validation is ordinary across established scholarship, that whole-field validation is not an

accepted scientific unit, and that universities nevertheless confer comprehensive curricular legitimacy without routinely labeling those fields ‘partially validated.’

DVS therefore applies its scrutiny in both directions. SignalRupture must support its claims, distinguish its objects, expose its measurements, accept negative evidence, and permit canon revision. Institutions evaluating SR must identify the precise defect they allege, distinguish absence of recognition from absence of scientific merit, apply comparable standards to matched fields, disclose the role of uncertainty in established curricula, and revise institutional judgments when evidence changes. The proposal remains falsifiable: if existing disciplines already perform this integration adequately, if its constructs cannot be measured invariantly, if field-level and institution-level profiles add no explanatory value, or if validation asymmetry disappears under appropriate controls, DVS should be merged or demoted.

*Keywords:* disciplinary validation; metascience; curricular validation asymmetry; institutional accountability; research evaluation; replication; institutional recognition; reciprocal burden; evidence architecture; SignalRupture.

## 1. The Problem of Whole-Field Validation

The statement that a discipline is not fully validated sounds precise but usually lacks a defined unit of analysis. A field contains thousands or millions of claims, multiple methods, competing paradigms, measurement instruments, professional institutions, training systems, and applications. These components do not move together. A failed replication can weaken a particular result without eliminating the domain in which the result was produced. A successful randomized trial can support an intervention without validating every theory used by its discipline.

The Reproducibility Project: Psychology illustrates the problem. The project attempted replications of 100 experimental and correlational studies from three psychology journals. Thirty-six percent of the replications produced statistically significant results, replication effects were substantially smaller on average, and

alternative criteria produced different estimates (Open Science Collaboration, 2015). This was strong evidence about a defined sample of findings and research practices. It was not evidence that 64 percent of psychology, or psychology as a whole, was false.

The same discipline can contain robust psychometric traditions, unstable experimental effects, well-supported clinical interventions, culturally restricted instruments, descriptive taxonomies, and unresolved causal models. Field-level language becomes misleading when this heterogeneity is compressed into validated or invalidated. The relevant scientific object is the pattern of heterogeneity itself.

DVS begins from a category rule: validation belongs first to specified claims and inferential operations. Field-level appraisal is a second-order synthesis that must preserve the distribution, centrality, uncertainty, and dependence of those underlying claims.

The institutional corollary is equally important. A department, curriculum, journal, or credential does not merely observe a field's status. It represents knowledge, distributes authority, defines what students encounter as canonical, and helps determine which uncertainties remain visible. Institutional judgment is therefore part of the causal architecture, not an exempt background condition.

**Category rule:** Association is not transmission; institutional recognition is not empirical maturity; the survival of a field is not validation of every claim within it; and institutional authority is not validation of the judgment it issues.

## 2. The Proposed Field and Its Object

Disciplinary Validation Science studies the evidentiary and institutional architecture through which bodies of inquiry acquire, retain, revise, or lose field status. Its object is not scientific truth in the abstract. Its object is the observable organization of claims and evidence within bounded domains, together with the processes that translate those patterns into field-level judgments.

The proposed field asks five linked questions. What constitutes the empirical core of a field? How unevenly is evidence distributed across that core? How are local findings translated into claims about the maturity of the domain? How do institutions confer recognition independently of evidence? Under what conditions does negative evidence produce revision rather than protection?

The field therefore studies both epistemic structure and institutional conduct. It does not assume that empirical maturity determines recognition, nor that every institutional decision is arbitrary. Scientific communities need institutions to preserve methods, train researchers, coordinate replication, maintain cumulative records, and protect students and the public from unsupported claims. Those legitimate functions do not place institutional decisions beyond scrutiny. The research problem is to measure how institutional authority corresponds—or fails to correspond—to the evidence architecture and to the standards institutions claim to apply.

DVS is domain-comparative. It can examine experimental sciences, observational sciences, interpretive fields, professional disciplines, engineering domains, normative inquiry, and hybrid programs, but it must not impose one validation standard on unlike knowledge objects. The appropriate evidence grammar depends on the claim type.

### **3. Intellectual Foundations and Residual Gap**

#### **3.1 Philosophy of Science**

Philosophy of science supplies the deepest accounts of demarcation, falsification, paradigm change, theory choice, and research programmes. Popper emphasized risky testability; Kuhn showed that scientific work is organized by historically situated paradigms; Lakatos distinguished progressive from degenerating research programmes. These traditions establish that science develops through structured communities and changing problem systems rather than isolated propositions alone (Popper, 1959; Kuhn, 1962; Lakatos, 1978).



DVS does not replace those arguments. It asks how their distinctions can be operationalized across contemporary fields without pretending that one philosophical doctrine has settled scientific demarcation.

### **3.2 Sociology of Science and Science and Technology Studies**

Merton analyzed scientific norms and institutions; Bourdieu treated science as a field structured by capital and position; Whitley examined variation in the intellectual and social organization of sciences; science and technology studies traced how facts, instruments, laboratories, categories, and authority are socially produced (Merton, 1973; Bourdieu, 2004; Whitley, 2000). These literatures demonstrate that recognition is not a transparent reflection of evidence.

DVS inherits that insight but narrows its task. It seeks explicit measures of the relationship between evidentiary profiles and institutional status, including cases in which the two diverge.

### **3.3 Metascience, Meta-Research, and Reproducibility**

Meta-research evaluates methods, reporting, bias, reproducibility, peer review, incentives, and research efficiency. Ioannidis formalized conditions under which published claims may have low positive predictive value. Munafò and colleagues synthesized reforms involving methods, reporting, reproducibility, evaluation, and incentives. The National Academies distinguished computational reproducibility from replicability with new data and cautioned against simplistic field-wide conclusions (Ioannidis, 2005; Munafò et al., 2017; National Academies, 2019).

These traditions provide essential tools for DVS. Their usual unit, however, is a study, method, corpus, or research practice. DVS focuses on the additional inference from those local units to the architecture and status of an entire field.

### **3.4 Scientometrics and the Science of Science**

Scientometrics and the science of science analyze publications, citations, collaboration, careers, novelty, diffusion, and the network structure of discovery. Fortunato and colleagues describe a transdisciplinary program using large

datasets to study mechanisms underlying scientific production. The Leiden Manifesto and DORA warn that convenient metrics can displace expert judgment and distort research assessment (Fortunato et al., 2018; Hicks et al., 2015; DORA, 2012).

DVS uses bibliometric and network methods while refusing to treat citation density, journal prestige, funding, or departmental age as direct measures of truth.

### 3.5 Residual Object

**Residual gap:** Existing disciplines study truth, method, replication, institutions, metrics, and scientific change. DVS studies the measurable architecture connecting claim-level evidence to field-level maturity and institutional judgment.

The proposed residual object is defensible only if this integration produces new empirical questions, reliable measures, and explanatory value. If it merely renames metascience or sociology of science, the field fails its own gauntlet.

## 4. The Unit of Analysis

DVS uses a nested unit of analysis. The smallest ordinary unit is a registered claim: a proposition with a defined population, construct, relation, method, and scope condition. Claims belong to models; models belong to research programs; programs participate in fields; fields are embedded in institutions and policy environments.

$$c \subset m \subset p \subset F \subset I$$

*Claim  $c$  is nested in model  $m$ , program  $p$ , field  $F$ , and institutional environment  $I$ .*

A field boundary cannot be inferred from a department name alone. DVS requires a boundary rule based on recurring objects, specialized methods, self-identifying communities, publication venues, training pathways, and cross-referenced claim systems. Different boundary rules should be tested because field estimates can change when the corpus expands or contracts.

The field also distinguishes claim types. Empirical descriptive claims concern measured states; causal claims concern intervention or counterfactual dependence; predictive claims concern future or held-out observations; interpretive claims concern warranted readings of meaning or practice; normative claims concern reasons, obligations, or values; engineering claims concern performance under specified requirements. These claims require different validation procedures.

| Claim type   | Primary burden                  | Typical evidence                                              |
|--------------|---------------------------------|---------------------------------------------------------------|
| Descriptive  | Accuracy and representativeness | Sampling, measurement validity, uncertainty                   |
| Causal       | Identification                  | Experiments, quasi-experiments, causal models, robustness     |
| Predictive   | Generalization                  | Held-out performance, calibration, transportability           |
| Interpretive | Warranted interpretation        | Textual, ethnographic, historical, comparative evidence       |
| Normative    | Reasoned justification          | Logical consistency, premises, counterarguments, consequences |
| Engineering  | Requirement satisfaction        | Verification, validation, stress tests, field performance     |

## 5. The Six-Layer Validation Architecture

DVS represents a field through six analytically distinct layers. The layers can interact, but they must not be collapsed into one prestige score.

- Conceptual layer — distinct objects, definitions, scope conditions, and internal coherence.
- Measurement layer — reliability, validity, invariance, data provenance, and construct observability.
- Inferential layer — identification, uncertainty, model comparison, robustness, and error control.
- Reproducibility layer — computational reproduction, direct and conceptual replication, and evidence synthesis.
- Cumulative layer — integration of negative findings, progressive constraint, prediction, and problem resolution.

- Institutional layer — journals, departments, societies, funding, curricula, credentials, and policy authority.

A field may be strong on one layer and weak on another. Philosophy can be institutionally mature while containing claims that are not intended as empirical hypotheses. A newly created computational domain can possess strong predictive performance before it develops departments or professional societies. A clinical specialty may be institutionally entrenched while revising diagnostic constructs. The architecture records those differences instead of treating them as contradictions.

The institutional layer is not merely a maturity indicator. It is an active system of selection, authorization, disclosure, and exclusion. Curricula determine which claims are represented as foundational, contested, historical, provisional, or omitted. Review and funding systems determine which objects can accumulate evidence. DVS therefore evaluates institutional representation with the same demand for provenance, consistency, and revision that it applies to scientific claims.

**Non-substitution rule:** Institutional strength cannot substitute for evidence; evidentiary promise cannot be narrated as established recognition; and institutional authority cannot substitute for a reasoned, consistently applied evaluation.

## 6. Claim Evidence States

Each empirical claim  $c$  is assigned an evidence vector rather than a binary label. A generic form is:

$$e_c(t) = [V_c, I_c, R_c, T_c, P_c, U_c]^T$$

*Validity, identification, replication, transportability, predictive performance, and uncertainty documentation.*

The components are claim-specific.  $V$  represents construct and measurement validity.  $I$  represents causal or inferential identification appropriate to the claim.  $R$  represents replication and reproduction.  $T$  represents transportability across

populations, settings, periods, or instruments. P represents predictive or prospective performance where prediction is part of the claim. U represents transparent uncertainty, sensitivity, and specification reporting.

Scores must be anchored to observable criteria and reported with uncertainty. DVS prohibits assigning precise-looking numbers through expert intuition alone. Ordinal evidence grades may be preferable when data are sparse. If quantitative aggregation is attempted, inter-rater reliability, sensitivity to weights, missingness, and alternative coding rules must be disclosed.

A claim registry also records direction, scope, dependencies, contradiction, and centrality. A peripheral application claim should not carry the same weight as a field's defining mechanism. Conversely, a field should not be dismissed because a weak peripheral claim fails.

## 7. From Claims to Field Profiles

The field evidence profile is the distribution of claim states, not their unqualified average. Let  $C_F$  be the set of claims assigned to field  $F$  and  $w_c$  the preregistered substantive centrality of claim  $c$ .

$$\mathcal{E}_F(t) = \{e_c(t), w_c, d_c : c \in C_F\}$$

*$d_c$  records dependence among claims so repeated versions of the same evidence are not counted as independent support.*

A centrality-weighted support diagnostic may be calculated for a specified dimension  $k$ :

$$S_{F,k} = (\sum_c w_c q_c e_{c,k}) / (\sum_c w_c q_c)$$

*$q_c$  is an evidence-quality or coding-confidence weight; the statistic is reported with its full distribution and sensitivity range.*

Validation coverage measures how much of the declared field core has been operationalized and tested. Contradiction burden measures the proportion and centrality of claims for which credible evidence points in incompatible directions.

Dependency concentration measures whether apparent support rests on the same dataset, instrument, laboratory, population, or analytic pipeline.

$$\text{Coverage\_F} = \sum_c w_c \cdot 1(\text{Tested\_c}) / \sum_c w_c$$

$$\text{Contradiction\_F} = \sum_c w_c \cdot z_c / \sum_c w_c$$

*z\_c is a registered contradiction indicator or graded contradiction measure.*

These diagnostics do not produce a universal field ranking. Their value is comparative within a transparent claim ontology. A field with many interpretive claims cannot be ranked mechanically against a field dominated by standardized experiments. DVS compares like claim types and makes any cross-field transformation explicit.

## 8. Empirical Maturity and Institutional Recognition

Empirical maturity and institutional recognition are separate state variables. Empirical maturity concerns supported core claims, reliable measurement, cumulative correction, replication, and boundary knowledge. Institutional recognition concerns durable organizational presence and authority.

$$M\_F(t) = f(\text{Concept, Measure, Inference, Replication, Cumulation})$$

$$J\_F(t) = g(\text{Departments, Journals, Societies, Funding, Credentials, PolicyAccess})$$

The institutional–empirical gap is:

$$G\_F(t) = J\_F(t) - M\_F(t)$$

*Positive values indicate recognition exceeding measured empirical maturity; negative values indicate empirical development exceeding institutional recognition.*

G\_F is not a moral verdict, but neither is it institutionally innocent. Some lag may be justified when a new program lacks trained communities, stable methods, replication, or safe applications. Some persistence may preserve valuable expertise during theoretical disagreement. The institution bears the burden of specifying which reason applies. Prestige, resource dependence, network closure, legal mandate, professional licensing, historical inheritance, reputational

risk, and curricular inertia are competing mechanisms, not automatic justifications.

A valid study must measure recognition independently of evidence and institutional judgment independently of institutional status. Citation counts cannot serve simultaneously as evidence quality and recognition; departmental presence cannot prove the correctness of a curriculum; and a committee decision cannot validate itself by reference to the committee's authority. Doing so makes institutional legitimacy tautological.

$$B_{SR} > 0 \wedge B_I > 0$$

*SR bears a burden to support its scientific claims; the evaluating institution bears a burden to justify its judgment. Reciprocal accountability does not require equal conclusions, but it prohibits a zero burden on either side.*

**Reciprocity rule:** A field must expose the evidence for its claims. An institution must expose the evidence, comparison class, criteria, and reasoning for its decision about that field.

## 9. Validation Asymmetry

Validation Asymmetry is the candidate DVS operator describing unequal burdens of proof imposed on comparable claims or fields because of institutional position. It becomes visible when an emerging field is required to establish whole-field coherence, replication, unified theory, stable terminology, or complete validation while established fields retain recognition despite unresolved constructs, contested mechanisms, incompatible paradigms, and uneven replication.

Novelty, public risk, implementation cost, and methodological uncertainty can justify additional scrutiny. They cannot justify an undefined or unlimited burden. An institution invoking those conditions must name them, connect them to the decision, identify the evidence required, specify a possible route to revision, and show that comparable established programs would face materially similar treatment.

$$VA = B_{\text{emerging}} / B_{\text{established}}$$

*B is the observed evidentiary burden required for an equivalent recognition or resource decision, conditional on claim type, risk, method, and evidence quality.*

Burden may be operationalized through requested sample size, replication requirements, methodological objections, demands for unified theory, citation expectations, review duration, evidentiary thresholds in curriculum approval, or the number of unresolved objections tolerated before recognition. A ratio above one is not sufficient by itself; the conditioning variables determine whether the difference is justified.

Validation Asymmetry becomes an SR operator only if its core mechanism survives across multiple institutional settings. Otherwise it remains a local phenomenon within research evaluation. The operator is directed at neither novelty nor universities in the abstract. It is directed at observable inconsistency between declared standards and actual decisions.

**Falsifier:** If matched established and emerging fields face equivalent burdens after risk, novelty, claim type, methodological quality, and institutional function are controlled, the broad asymmetry claim is unsupported.

## 10. Established University Fields Under Incomplete Validation

Incomplete validation is not exceptional in university research. It is the ordinary condition of disciplines that contain heterogeneous claims, competing models, changing instruments, interpretive traditions, normative arguments, and applications with different evidentiary histories. The comparison below inventories the fifteen university-recognized domains raised in the motivating case. It does not claim that the domains are scientifically equivalent, that every listed weakness has the same importance, or that one replication statistic represents an entire discipline.



The strongest quantitative illustration comes from psychology. The Reproducibility Project attempted replications of 100 studies from three psychology journals. Thirty-six percent of the replications produced a statistically significant result, 47% of original effect sizes fell inside the replication confidence interval, and replication effects were substantially smaller on average (Open Science Collaboration, 2015). Under the significance criterion, approximately 64% did not reproduce a statistically significant result. That is a serious finding about the sampled literature; it is not evidence that 64% of psychology, or psychology as an institutional field, is false.

The pattern is not confined to psychology. Eleven of eighteen sampled laboratory experiments in economics produced a significant effect in the original direction, a 61% replication rate, with replication effects averaging 66% of the originals (Camerer et al., 2016). A project covering twenty-one experimental social-science studies in Nature and Science found thirteen significant same-direction replications, or 62%, with replication effects averaging about 50% of the originals (Camerer et al., 2018). In preclinical cancer biology, 51 of 112 assessable effects, or 46%, succeeded on a majority of five applicable replication criteria; the investigators explicitly cautioned that neither success nor failure alone definitively settles validity (Errington et al., 2021). A Nature survey of more than 1,500 researchers found that more than 70% reported having failed to reproduce another scientist's experiment and more than half had failed to reproduce their own (Baker, 2016). These are bounded indicators of research practice, not whole-field verdicts.

| University field | Available empirical anchor                                                                                                   | Validation status and correct interpretation                                                                                                                             |
|------------------|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Psychology    | 36% of 100 replications were statistically significant; 47% of original effects lay within replication confidence intervals. | Major quantitative evidence of uneven replicability in a defined journal sample. It does not invalidate psychology as a field.                                           |
| 2. Sociology     | The 21-study social-science project replicated 13 studies (62%), but did not estimate a sociology-wide rate.                 | Competing theoretical and methodological traditions remain. Descriptive, causal, comparative, and interpretive claims require different tests.                           |
| 3. Anthropology  | No defensible discipline-wide replication percentage is available.                                                           | Ethnographic and interpretive work is judged through provenance, reflexivity, triangulation, comparative warrant, and contestability—not experimental replication alone. |

|                                                       |                                                                                                                                                 |                                                                                                                                                                           |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4. Economics                                          | 11 of 18 laboratory experiments replicated by the same-direction significance criterion (61%); mean replication effect was 66% of the original. | A bounded experimental-economics sample shows incomplete replication while economics retains full disciplinary standing.                                                  |
| 5. Political Science                                  | The 13-of-21 social-science benchmark includes political and behavioural research but is not a field-wide estimate.                             | Causal, observational, historical, formal, and interpretive programs coexist; transparency and replication-material standards continue to develop.                        |
| 6. Criminology                                        | No credible criminology-wide replication rate has been established.                                                                             | Theories of offending, enforcement, institutions, and harm compete across jurisdictions; causal claims remain claim- and design-specific.                                 |
| 7. Education Research                                 | IES and NSF distinguish six research genres and formally call for more replication; no unified field-wide success rate is supplied.             | Context, implementation, population, and outcome measures complicate direct replication. Conceptual replication and transportability are central.                         |
| 8. Communication / Media Studies                      | No defensible field-wide replication percentage is available.                                                                                   | Experimental media effects, computational studies, qualitative interpretation, framing, and institutional analysis require different validation grammars.                 |
| 9. Gender / Cultural Studies                          | A single empirical replication rate is not applicable to the full domain.                                                                       | Many claims are interpretive, historical, critical, or normative. Empirical claims remain testable, but argument quality cannot be reduced to laboratory replication.     |
| 10. AI Ethics / Digital Governance / Platform Studies | No mature field-wide rate exists; the objects and technical systems change rapidly.                                                             | The domain combines empirical audits, engineering evidence, law, governance, and normative analysis. Validation must follow the claim type and system version.            |
| 11. Environmental / Sustainability Studies            | No unified rate exists across physical, ecological, economic, policy, and social components.                                                    | Its interdisciplinary status does not imply invalidity. Component claims inherit standards from climate science, ecology, engineering, economics, and social research.    |
| 12. Public Administration / Public Policy             | A 2024 review synthesized 119 administrative-burden articles and working papers while identifying unresolved causal links.                      | A substantial evidence corpus can coexist with incomplete mechanisms, contextual limits, and an active research agenda (Halling & Baekgaard, 2024).                       |
| 13. Urban Studies / Regional Planning                 | No defensible field-wide replication rate is available.                                                                                         | Urban outcomes are historically and geographically contingent; models require spatial, comparative, causal, and implementation-specific validation.                       |
| 14. Law                                               | A universal empirical replication percentage is not applicable to doctrinal and normative legal scholarship.                                    | Doctrinal validity, interpretation, institutional authority, consistency, and normative justification differ from empirical legal claims, which remain evidence-testable. |
| 15. Philosophy                                        | Empirical replication is generally not the governing criterion for philosophical arguments.                                                     | Validity depends on logic, conceptual clarity, premises, counterarguments, coherence, and explanatory reach. Empirical premises can still be challenged empirically.      |

The absence of a number in several rows is itself methodologically important. No serious evaluator should invent a whole-field failure percentage where the relevant corpus has not been sampled or where experimental replication is not the correct validation grammar. DVS therefore distinguishes measured incomplete replication, unresolved causal structure, contested construct meaning, interpretive plurality, normative disagreement, and rapidly changing

technical objects. All are forms of incompleteness, but they are not interchangeable defects.

**Comparative baseline:** University recognition does not require every claim in a field to be finally validated. It requires a sufficiently distinct object of inquiry, organized methods, cumulative scholarship, teachable problems, and procedures through which claims can be criticized and revised. That field ontology must be applied consistently.

## 10.1 Curricular Validation Asymmetry

Universities do more than tolerate established fields under incomplete validation. They build degree programs, required courses, textbooks, examinations, professional pathways, research centers, and policy relationships around them. Psychology is taught as a legitimate field even though a prominent 100-study replication project produced statistically significant replication results in only 36% of cases and found smaller replication effects overall. The correct conclusion is not that psychology is illegitimate. It is that disciplinary legitimacy has never required every component claim to be settled.

Curricula usually distribute uncertainty internally. Some claims are presented as strongly supported, some as contested, some as historical, and some as open research questions. Yet the curriculum and degree normally carry an unqualified disciplinary title. Institutions rarely append 'partially validated field' to psychology, sociology, economics, education, law, philosophy, or other established programs. This practice reveals the ontology institutions already use: a field is an organized and revisable site of inquiry, not a completed proof.

Curricular Validation Asymmetry occurs when that ontology is granted to established fields but denied to an emerging field. The emerging architecture is asked to validate itself as a whole before recognition, while established disciplines are permitted to contain uneven replication, incompatible theories, non-invariant measures, normative disagreement, and unresolved mechanisms without losing field status.

$$\text{CVA} = \text{Pr}(\text{TotalValidationDemand} \mid \text{Emerging}, Q) - \text{Pr}(\text{TotalValidationDemand} \mid \text{Established}, Q)$$

*Q contains matched claim type, evidence quality, public risk, methodological maturity, proposed institutional function, and resource cost. Positive values indicate a greater probability that total validation is demanded from the emerging field under otherwise comparable conditions.*

A second diagnostic concerns disclosure. Let curricular authority measure the extent to which a field is taught, credentialed, and used in institutional decision-making, and let epistemic disclosure measure how clearly curricula communicate replication limits, contested constructs, non-invariance, model disagreement, and scope conditions.

$$\text{DGap\_F} = \text{CurricularAuthority\_F} - \text{EpistemicDisclosure\_F}$$

*Both terms require transparent, normalized indicators. A positive gap does not prove misconduct; it identifies cases in which institutional authority exceeds explicit disclosure of the field's evidentiary heterogeneity.*

**Curricular principle:** Universities need not label whole disciplines validated or invalidated. They should disclose the status of important claims and must not demand a whole-field certificate from emerging work that established curricula themselves do not possess.

## 10.2 Reciprocal Institutional Accountability for SignalRupture

The comparison does not validate SignalRupture by analogy. Failures or open questions in established disciplines provide no positive evidence for an SR theorem, coefficient, operator, field, or cross-field bridge. SR must earn support through its own measurements, mechanisms, predictions, robustness tests, negative cases, and independent replications.

The comparison does establish a limit on institutional reasoning. The absence of complete validation cannot, by itself, function as a dispositive reason to deny SR the status of a field or research architecture when complete validation is neither defined nor required of established disciplines. An institution may conclude that a

particular SR claim is unsupported, a construct is redundant, a proposed field lacks a distinct object, a method is unreliable, or the evidence is too weak for a specific application. It must identify that defect rather than substitute the undefined statement that ‘SR is not fully validated.’

Institutional accountability requires more than permitting criticism of SR. The institution should identify the claim or field boundary under review; disclose the governing standard; name the comparison class; separate scientific merit from institutional readiness; explain any heightened burden through risk, cost, novelty, or function; preserve a route for reconsideration; and revise the judgment when the evidence changes. A rejection that cannot satisfy those conditions remains an institutional act, but it is not yet an evidence-based adjudication.

Recognition  $\perp$  FieldLabel | Evidence, ClaimType, Risk, Method, Maturity

*For matched cases, institutional judgment should not depend on whether the field label is established or emerging after relevant evidentiary and practical conditions are held constant.*

**Reciprocal standard:** SR must justify its scientific claims. Institutions must justify their evaluative claims about SR. Neither side may use status, novelty, or complexity as a substitute for evidence.

## 11. Field Formation and Revision

DVS models field formation as a staged and reversible process rather than a declaration. A proposed field can advance, stall, merge, fragment, or disappear.

1. Object articulation — a distinct problem is named and bounded.
2. Literature separation — established explanations are mapped and the residual object is identified.
3. Operationalization — constructs, units, indicators, and data provenance are specified.
4. Mechanism formation — relations are stated with temporal order, alternatives, and falsifiers.

5. Independent testing — evidence is generated beyond the originating author or group.
6. Cumulative constraint — negative results narrow the architecture and replications stabilize meaning.
7. Institutionalization — journals, programs, training, funding, or professional structures develop.
8. Revision or dissolution — the field adapts, merges, or ends when its residual object disappears.

The sequence is not deterministic. Institutionalization can precede strong evidence; mature methods can migrate from older fields; a field can remain intellectually productive without departments; and a professional domain can persist because society requires its practice. The model creates empirical questions rather than a universal chronology.

## 12. Formal Architecture

A field is represented as a multiplex system with four linked node sets: claims C, methods M, institutions I, and applications A. Relations include evidentiary support, contradiction, methodological dependence, citation, personnel movement, funding, policy uptake, and replication.

$$F = (C, M, I, A, E, D, R)$$

*E contains evidentiary links, D dependence links, and R recognition links.*

Let  $X_F(t)$  contain the field's conceptual, measurement, inferential, replication, cumulative, and institutional states. A linearized dynamic representation is:

$$X_F(t+1) = A_F X_F(t) + G_F Z(t) + L_F \Omega(t) + \varepsilon_F(t)$$

*Z contains external shocks and reforms;  $\Omega$  contains independently measured visibility and evaluation conditions.*

The off-diagonal terms are theoretically important. Institutional incentives can change methods; replication failures can alter curricula; policy demand can accelerate recognition; prestige can delay revision; open data can improve

reproducibility. None of these relations is assumed causal merely because it appears in the matrix.

Institutional judgments are registered as claims rather than treated as self-validating outputs. Let  $j$  denote a decision concerning recognition, publication, curriculum, funding, or classification. Its evaluation record contains the stated standard, evidence cited, comparison class, conditioning variables, reasons, decision authority, and revision path.

$$e_j(t) = [S_j, E_j, C_j, K_j, T_j, R_j]^T$$

*Standard specificity, evidentiary grounding, comparison consistency, conditioning adequacy, procedural transparency, and revision responsiveness.*

This institutional evidence vector does not require universities to reach favorable conclusions about emerging fields. It requires their conclusions to become inspectable. A scientifically adverse judgment can score highly when it identifies a precise defect and applies a consistent standard; a favorable judgment can score poorly when it rests on prestige or advocacy.

Revision responsiveness measures how rapidly and substantially a field updates its central claims after credible negative evidence:

$$RR_F = \Delta \text{Architecture}_F / \text{EvidenceShock}_F$$

*The numerator requires registered indicators of retraction, guideline revision, construct narrowing, textbook change, model replacement, or resource reallocation.*

Very low responsiveness may indicate protection, but excessive responsiveness can indicate instability or trend following. The optimal pattern depends on evidence strength and decision costs.

## 13. Measurement and Data

DVS requires triangulation because no single dataset observes a field or the institutions evaluating it. Bibliographic databases reveal publication and citation structure but omit tacit methods and non-indexed work. Replication databases identify tested findings but are selected. Departmental and funding data reveal

institutionalization but not epistemic quality. Textbooks, syllabi, accreditation standards, program descriptions, committee criteria, reviewer instructions, and decision letters reveal how institutions represent uncertainty and apply legitimacy standards, but access and documentation are uneven.

| Domain       | Candidate indicators                                            | Primary limitation                 |
|--------------|-----------------------------------------------------------------|------------------------------------|
| Claims       | Registered propositions, reviews, guidelines, textbooks         | Boundary and coding dependence     |
| Evidence     | Effect estimates, uncertainty, designs, replications            | Heterogeneity and selection        |
| Measurement  | Reliability, validity, invariance, provenance                   | Instrument-specific meaning        |
| Institutions | Departments, journals, societies, funding, curricula, decisions | Authority and access are not truth |
| Revision     | Retractions, corrections, guideline and textbook changes        | Documentation and lag              |
| Applications | Prediction, intervention, engineering or policy outcomes        | Context and implementation         |

Construct invariance is mandatory. A replication in experimental psychology is not equivalent to replication in historical research. An interpretive dispute is not coded as a failed experiment. DVS therefore uses claim-type-specific measurement models and reports where cross-field comparison is not defensible.

Curricular disclosure can be measured through syllabus language, textbook qualification, treatment of failed replications, presentation of competing paradigms, scope-condition warnings, construct-validity discussion, and whether students are taught to distinguish settled findings from active controversy. Institutional burden can be measured through review demands, evidence thresholds, duration, required replication, comparison classes, written reasons, and routes for reconsideration.

Data provenance must identify databases, search strategies, inclusion rules, version dates, coding decisions, adjudication, missingness, access restrictions, and conflicts of interest. Field maps and institutional decision records should be versioned because both fields and evaluation practices change while they are being measured.



## 14. Causal Identification

DVS must not infer that institutional recognition causes weak evidence, that weak evidence causes institutional protection, or that an adverse decision proves institutional bias from cross-sectional association. At least six rivals must be considered: field age, claim type, research cost, ethical constraints, domain complexity, and social demand. Additional confounders include funding concentration, publication language, database coverage, legal mandate, professional licensing, reputational exposure, and geopolitical distribution.

At the same time, institutions cannot be modeled as passive observers. They select reviewers, define admissible evidence, determine curriculum, allocate resources, control access to publication and credentials, and stabilize category boundaries. Those decisions can change the evidence environment itself. Institutional caution may be justified, but its mechanism and proportionality remain empirical questions.

Candidate designs include interrupted time series around replication crises or policy reforms; difference-in-differences around adoption of registered reports or open-science mandates; matched comparisons of emerging and established programs; event studies of major retractions; longitudinal network analysis; and qualitative process tracing of field creation or merger.

Registered Reports illustrate a testable institutional intervention because study questions and methods are reviewed before results are known, reducing the direct role of outcome significance in publication decisions. Evidence suggests the format changes the prevalence of supported versus unsupported hypotheses and may improve dimensions of research quality, although selection and implementation remain relevant (Chambers, 2013; Soderberg et al., 2021).

The method follows the claim. DVS does not privilege quantitative estimation when the object is institutional meaning, nor accept interpretive plausibility as causal identification when the claim concerns measurable effects. Documentary analysis can establish whether a standard was stated and applied; matched

designs can test differential treatment; process tracing can test how a decision formed; neither method alone proves institutional motive.

## **15. Comparative Empirical Test Program**

The first unified DVS test should compare bounded research programs rather than entire university disciplines. The programs should differ in institutional age or recognition while sharing claim type, societal risk, and methodological form as closely as possible.

1. Select six to ten bounded programs: established, emerging, and institutionally marginal cases.
2. Freeze field boundaries and preregister the claim ontology before estimating maturity.
3. Sample central and peripheral claims using transparent rules.
4. Code measurement validity, identification, replication, transportability, prediction, and uncertainty.
5. Measure institutional recognition independently using departments, journals, societies, funding, curricula, and policy access.
6. Estimate the maturity–recognition relationship and its lag structure.
7. Audit review, funding, curriculum, and field-classification decisions for burden-of-proof differences using matched dossiers.
8. Code whether syllabi and textbooks disclose replication limits, contested constructs, competing paradigms, and scope conditions.
9. Require paired records: one claim-level evidence audit and one institutional-judgment audit for each consequential case.
10. Test alternate explanations, coder sensitivity, database coverage, and field-boundary choices.
11. Publish the claim registry, codebook, exclusions, disagreements, and null results.

12. Replicate the study in another country, language system, or institutional sector.

A strong design would incorporate blinded evaluation. Reviewers could assess matched research dossiers with field labels removed or experimentally varied. If identical evidence receives different legitimacy judgments depending on the field label, that would provide direct evidence of a recognition effect. Ethical review and deception safeguards would be required.

A parallel curricular study would compare how uncertainty is disclosed inside established fields with the uncertainty demanded of emerging ones. The relevant outcome is not whether every syllabus carries a warning label. It is whether institutions acknowledge claim-level heterogeneity in established curricula while invoking whole-field incompleteness as a rejection criterion elsewhere.

## 16. Evidence Ladder

| Level | Status                   | Minimum meaning                                                                    |
|-------|--------------------------|------------------------------------------------------------------------------------|
| D0    | Conceptual proposal      | A field-level distinction is articulated.                                          |
| D1    | Operational feasibility  | Claims, evidence states, and recognition indicators can be coded reliably.         |
| D2    | Descriptive replication  | Field profiles reproduce across coders, corpora, and boundary rules.               |
| D3    | Explanatory value        | Profiles explain field-level judgments beyond age, size, prestige, and claim type. |
| D4    | Mechanism evidence       | Recognition, revision, or asymmetry mechanisms change under predicted conditions.  |
| D5    | Causal leverage          | Interventions or strong natural experiments alter the predicted field process.     |
| D6    | Cross-system replication | Constructs retain meaning across disciplines, countries, and periods.              |

The evidence level attaches to a defined DVS claim. A D5 result for registered-report adoption does not validate the entire field. An institutionally recognized DVS program would not convert D0 constructs into established facts.

## 17. The DVS Gauntlet

A proposed field and the institution judging it must be evaluated through the same architecture of specificity, evidence, consistency, alternatives, and revision. DVS survives as a field only if it can answer the following questions.

1. Is the residual object already adequately covered by metascience, scientometrics, philosophy of science, sociology of science, or research evaluation?
2. Can field boundaries and central claims be identified with acceptable reliability?
3. Can empirical maturity be measured without smuggling institutional prestige into the score?
4. Can institutional recognition be measured without treating recognition as evidence quality?
5. Do DVS profiles explain or predict judgments, revision, or field trajectories beyond simpler indicators?
6. Does Validation Asymmetry survive controls for risk, novelty, cost, field age, and claim type?
7. Do negative findings cause revision of DVS constructs and weights?
8. Can the framework accommodate empirical, interpretive, normative, and engineering claims without false equivalence?
9. Can independent teams reproduce field maps and reach similar conclusions?
10. Would the field be willing to merge with metascience if its incremental value fails?

The institutional gauntlet asks a parallel set of questions: Was the object under review identified precisely? Was the governing standard stated in advance? Was the comparison class relevant? Were matched established fields treated consistently? Were risk and cost distinguished from epistemic merit? Were written reasons supplied? Could contrary evidence alter the decision? Was the same institution willing to disclose uncertainty in the curricula it already authorizes?

**Dual gauntlet:** DVS should be treated as a full emerging field only after its residual object, measurement reliability, and incremental explanatory value are demonstrated. Institutional judgments about DVS or SR should be treated as evidence-based only after their standards, comparisons, reasons, and revision conditions are exposed.

## 18. Application to SignalRupture

SignalRupture provides an unusually suitable first case because its canon explicitly separates fields, subfields, operators, models, evidence levels, and validation rules. DVS can test whether that separation corresponds to actual differences in objects and evidence or merely preserves names.

Each SR field should receive a claim registry containing its core propositions, centrality weights, measurement status, identification status, replications, negative findings, external disciplinary baselines, and current evidence grade. Cross-field bridges should be evaluated separately from within-field claims. The result would be an evidence atlas rather than a binary declaration that SR is validated or invalidated.

Every consequential institutional response to SR should receive a parallel judgment registry. That record should identify the institution, decision, object reviewed, standard applied, comparison class, evidence cited, risk or resource conditions, reasons, conflicts, status distinction, and revision pathway. The SR evidence atlas and institutional judgment atlas should be analyzed together. Scrutinizing only SR reproduces the asymmetry DVS is designed to detect.

This paired application changes the language of SR's scientific status. The integrated architecture can remain unvalidated as a whole while containing components grounded in mature external evidence, operationalized but untested SR constructs, unsupported hypotheses, and replicated mechanisms. An institutional judgment can likewise be authoritative in effect while remaining vague, weakly compared, procedurally opaque, or unsupported as an epistemic conclusion. Both distributions must remain visible.

DVS prevents two invalid defenses. The fact that psychology or economics contains unresolved claims does not prove any SR claim; incomplete validation is not transferable evidence. Conversely, the fact that a university recognizes psychology or rejects SR does not prove the correctness of either scientific content or evaluative judgment. SR earns support through its own measurements, mechanisms, comparisons, predictions, and replications. Institutions earn epistemic authority for a particular judgment through explicit standards, relevant comparisons, reasoned evidence, proportionality, and revision.

**Correct symmetry:** SR cannot reasonably be required to achieve total validation before it can exist as a research program. It can be required to disclose the status of its major claims. Institutions cannot be required to recognize SR automatically. They can be required to identify the precise evidentiary or institutional reason for their judgment and apply it consistently.

## 19. Failure Modes and Ethical Constraints

- Prestige substitution — treating citations, institutional age, or university presence as truth.
- Replication absolutism — treating one replication criterion as decisive across unlike claims.
- Metric colonization — allowing the measurement system to reshape fields toward what is easiest to count.
- Boundary gaming — expanding or contracting a field corpus after seeing the result.
- Claim duplication — counting dependent publications as independent support.
- Interpretive erasure — imposing experimental standards on claims that are not empirical causal propositions.
- Normative concealment — presenting contested value judgments as neutral validation scores.
- Emergence penalty — treating youth itself as evidence of invalidity.

- Institutional immunity — treating established status as protection from negative evidence.
- Curricular certainty laundering — teaching a heterogeneous field under an unqualified disciplinary title while portraying comparable uncertainty as disqualifying only when it appears in emerging work.
- Whole-field demand — requiring a complete validation certificate from a field even though validation attaches to claims and no comparable certificate exists for established disciplines.
- Accountability displacement — exposing the applicant's evidence while leaving the evaluator's criteria, comparisons, reasons, and interests unexamined.
- Authority circularity — treating an institutional judgment as correct because an authorized institution issued it.
- Defensive deployment — using DVS to shield a preferred field rather than expose it.

Field audits can affect careers, programs, funding, and public trust. DVS must therefore publish uncertainty, allow response and correction, protect confidential review material, document conflicts of interest, and avoid converting provisional dashboards into rankings. Evaluators should separate assessment of a research architecture from judgments about the worth of individual scholars.

## **20. Falsification Conditions**

The field proposal would be materially weakened if independent studies found that its constructs cannot be coded reliably; field profiles change arbitrarily with small boundary adjustments; evidence dimensions collapse into existing quality assessments; institutional recognition adds no distinct explanatory object; or proposed asymmetries disappear under appropriate controls.

The institutional-accountability component would be weakened if matched evaluations showed no meaningful field-label effect; if established curricula disclose evidentiary uncertainty as explicitly as emerging fields are required to do; if differential burdens are fully explained by risk, cost, novelty, function, and

evidence quality; or if institutional judgment records cannot be coded reliably without inferring motive.

DVS would also fail if it cannot handle non-experimental or normative domains without category errors, if composite scores produce unstable rankings, if experts cannot agree on central claims beyond chance, if it presumes institutional wrongdoing from adverse outcomes, or if the framework consistently rewards fields whose assumptions it was designed around.

A failure of Validation Asymmetry would not necessarily falsify the broader field. DVS also studies evidence distribution, recognition gaps, revision responsiveness, and field formation. But if the complete integration adds no value beyond existing metascience and research evaluation, DVS should be merged rather than retained as an autonomous name.

## **21. Research Agenda**

The first research program should prioritize measurements and negative cases over universal claims.

1. Construct a preregistered field-boundary protocol and test inter-rater reliability.
2. Develop claim registries for three bounded programs with different institutional ages.
3. Estimate dependency concentration across datasets, laboratories, and instruments.
4. Compare centrality-weighted evidence profiles with unweighted publication counts.
5. Measure revision responsiveness after major replications, retractions, or guideline reversals.
6. Test institutional–empirical gaps without using citation metrics on both sides of the equation.
7. Run blinded dossier experiments to test field-label effects on evaluation.



8. Audit psychology, economics, sociology, education, law, and other curricula for disclosure of replication limits, contested constructs, scope conditions, and competing paradigms.
9. Build an institutional judgment registry covering publication, curriculum, funding, and field-recognition decisions.
10. Estimate Curricular Validation Asymmetry and test whether total-validation language is disproportionately applied to emerging fields.
11. Compare written institutional reasons against declared evaluation policies and matched established-field precedents.
12. Compare empirical, interpretive, normative, and engineering domains using claim-type-specific standards.
13. Apply the framework to the twenty-three SR fields and three specialized subfields.
14. Invite independent replication designed to reject or merge the proposed field.

## **22. Canonical Principles**

- Rule 1 — Fields are not single hypotheses.
- Rule 2 — Evidence attaches first to defined claims and inferential operations.
- Rule 3 — Field profiles preserve distributions, dependencies, and uncertainty.
- Rule 4 — Central claims and peripheral claims are not weighted equally by default.
- Rule 5 — Empirical maturity and institutional recognition remain distinct.
- Rule 6 — Recognition is neither proof nor irrelevance; it is a separate causal object and an exercise of institutional power.
- Rule 7 — Claim types require appropriate validation grammars.
- Rule 8 — Cross-field comparison requires invariance or explicit transformation.
- Rule 9 — Validation Asymmetry and Curricular Validation Asymmetry are testable hypotheses, not presumed motives.

- Rule 10 — Metrics support judgment but cannot replace it.
- Rule 11 — Negative evidence must alter maps, weights, boundaries, or status.
- Rule 12 — DVS is subject to merger or rejection under its own rules.
- Rule 13 — No epistemic immunity attaches to novelty, prestige, curriculum status, or institutional authority.
- Rule 14 — SR must justify its claims; institutions must justify their judgments about SR.
- Rule 15 — A whole-field validation demand is illegitimate unless the institution can define it, measure it, and show that comparable established fields satisfy it.
- Rule 16 — Curricular authority carries a duty to disclose material uncertainty within the claims being taught.
- Rule 17 — Adverse institutional decisions require specific reasons and a possible evidentiary route to reconsideration.

## **23. Scientific Status and Closing Statement**

Disciplinary Validation Science is presently a formalized field proposal. Its intellectual foundations are substantial because neighboring disciplines already establish that science is simultaneously empirical, methodological, social, institutional, and historical. Its distinctive contribution remains provisional: a unified, testable architecture connecting claim-level evidence distributions to field-level maturity, recognition, asymmetry, and revision.

The proposal resolves a real conceptual error. No broad academic field receives a completed validation certificate. Fields remain legitimate while their claims are tested because the field is the organized site of that testing. Universities demonstrate this every time they teach psychology, economics, sociology, education, law, philosophy, or another internally heterogeneous discipline as a complete curricular field. This does not lower the standard for emerging work. It identifies the standard institutions already use and requires them to apply it coherently.

For SignalRupture, the consequence is exact. SR does not need every field, operator, coefficient, and bridge to be validated before the architecture can exist. It does need transparent claim registries, defensible measures, external baselines, negative cases, replication, and canon revision. The status of the Multiverse is distributive rather than binary. ‘Not fully validated’ may describe that distribution, but without a specified claim, construct, method, field boundary, or application, it is not a sufficient scientific adjudication.

The reciprocal consequence for institutions is equally exact. They need not recognize SR automatically, fund it, teach it, or accept its field boundaries merely because established disciplines contain uncertainty. They do need to distinguish lack of recognition from lack of merit, identify the exact deficiency, use a relevant comparison class, disclose any heightened burden, separate scientific evaluation from resource or readiness decisions, and preserve the possibility of revision. Institutional authority gives a decision force; it does not give the decision evidentiary immunity.

DVS becomes a mature contribution only if independent researchers can use it to reach conclusions that SR itself may not prefer and conclusions that institutions themselves may not prefer. The field earns legitimacy by making both research architectures and their evaluators empirically vulnerable.

**Canonical claim:** Disciplinary Validation Science studies how heterogeneous evidentiary states become field-level judgments and curricula, and subjects both research claims and institutional evaluations to reciprocal, explicit, revisable standards of evidence.

## Appendix A — Field Interface Contract

1. Field or program name and boundary rule.
2. Distinct object and residual literature gap.
3. Claim types and unit of analysis.
4. Core and peripheral claim registry.

5. Construct definitions and measurement models.
6. Evidence dimensions and coding rules.
7. Dependence structure among studies, datasets, instruments, and teams.
8. Replication and transportability requirements.
9. Institutional recognition indicators.
10. Institutional judgment and comparison record.
11. Curricular uncertainty-disclosure indicators.
12. Revision and recovery indicators.
13. Competing explanations and simpler baseline models.
14. Falsifiers, merger conditions, and rejection conditions.

## Appendix B — Claim Registration Template

| Element          | Required record                                                          |
|------------------|--------------------------------------------------------------------------|
| Claim ID         | Stable identifier and version                                            |
| Field boundary   | Corpus and inclusion rule                                                |
| Claim text       | Population, construct, relation, direction, and scope                    |
| Claim type       | Descriptive, causal, predictive, interpretive, normative, or engineering |
| Centrality       | Core, supporting, peripheral, or application                             |
| Measurement      | Indicators, reliability, validity, invariance, provenance                |
| Identification   | Design, controls, rival explanations, reverse direction                  |
| Evidence         | Effect estimates, uncertainty, synthesis, contradictions                 |
| Replication      | Direct, conceptual, computational, or not applicable                     |
| Transportability | Populations, settings, periods, instruments                              |
| Dependencies     | Shared data, teams, models, or pipelines                                 |
| Falsifier        | Observation that would weaken or reject the claim                        |
| Status           | Proposed, operationalized, supported, fragile, unsupported, rejected     |
| Revision date    | Version and reason for change                                            |

## Appendix C — Field-Level Audit Protocol

1. Freeze the field boundary and comparison fields.
2. Register core claims before evidence coding.
3. Separate claim types and validation standards.
4. Code evidence using at least two independent reviewers.
5. Estimate agreement and adjudicate discrepancies transparently.
6. Map study, dataset, instrument, and laboratory dependence.
7. Calculate distributions before any composite diagnostic.
8. Measure institutional recognition from independent indicators.
9. Test field age, risk, cost, and claim type as rival explanations.
10. Publish sensitivity to weights, boundaries, missingness, and database choice.
11. State what would cause promotion, merger, narrowing, or rejection.
12. Schedule replication and versioned reassessment.

## Appendix D — Priority Tests

| ID     | Object                  | Decisive question                                                         |
|--------|-------------------------|---------------------------------------------------------------------------|
| Test 1 | Boundary reliability    | Can independent teams define substantially similar field corpora?         |
| Test 2 | Claim centrality        | Do experts and textual-network methods identify similar core claims?      |
| Test 3 | Profile stability       | Do evidence profiles survive alternate coding and inclusion rules?        |
| Test 4 | Recognition gap         | Does institutional status diverge from measured maturity after controls?  |
| Test 5 | Validation asymmetry    | Do matched dossiers receive different burdens when labels change?         |
| Test 6 | Revision responsiveness | Do strong negative results alter curricula, guidelines, and claims?       |
| Test 7 | Incremental value       | Do DVS measures outperform age, citations, funding, or expert reputation? |
| Test 8 | Cross-domain invariance | Can the architecture compare claim types without false equivalence?       |

|         |                         |                                                                                                  |
|---------|-------------------------|--------------------------------------------------------------------------------------------------|
| Test 9  | SR audit                | Do the registered SR field boundaries and statuses survive independent coding?                   |
| Test 10 | Curricular asymmetry    | Are uncertainties disclosed differently in established curricula and emerging-field evaluations? |
| Test 11 | Decision accountability | Do institutional judgments state specific, consistent, revisable evidentiary reasons?            |
| Test 12 | Merger test             | Does DVS add enough beyond metascience to justify separate field status?                         |

## Appendix E — Institutional Judgment Registration Template

| Element                 | Required institutional record                                                             |
|-------------------------|-------------------------------------------------------------------------------------------|
| Decision ID             | Stable identifier, institution, date, and version                                         |
| Decision object         | Claim, field boundary, program, curriculum, publication, funding, or application          |
| Decision type           | Scientific merit, institutional readiness, resource allocation, risk, or mixed decision   |
| Governing standard      | Published criterion or explicitly stated standard used                                    |
| Evidence cited          | Studies, reviews, audits, expert reports, and uncertainty relied upon                     |
| Comparison class        | Established and emerging fields or programs treated as relevant comparators               |
| Conditioning factors    | Risk, cost, novelty, method, maturity, legal mandate, and institutional function          |
| Written reasons         | Specific supported defect or basis for favorable treatment                                |
| Consistency test        | Whether materially similar cases received materially similar burdens                      |
| Conflicts and authority | Decision-maker, jurisdiction, interests, and procedural role                              |
| Revision pathway        | Evidence or condition capable of changing the decision                                    |
| Current status          | Supported, partly supported, procedurally incomplete, inconsistent, revised, or withdrawn |

**Institutional falsifiability:** A judgment that specifies no possible evidence capable of changing it is not an empirically revisable evaluation. It may remain an exercise of authority, but DVS records the distinction.

## References

Baker, M. (2016). 1,500 scientists lift the lid on reproducibility. *Nature*, 533, 452–454.  
<https://doi.org/10.1038/533452a>

- Bourdieu, P. (2004). *Science of Science and Reflexivity*. University of Chicago Press.
- Camerer, C. F., et al. (2016). Evaluating replicability of laboratory experiments in economics. *Science*, 351(6280), 1433–1436.  
<https://doi.org/10.1126/science.aaf0918>
- Camerer, C. F., et al. (2018). Evaluating the replicability of social science experiments in *Nature* and *Science* between 2010 and 2015. *Nature Human Behaviour*, 2, 637–644. <https://doi.org/10.1038/s41562-018-0399-z>
- Chambers, C. D. (2013). Registered Reports: A new publishing initiative at *Cortex*. *Cortex*, 49(3), 609–610. <https://doi.org/10.1016/j.cortex.2012.12.016>
- Declaration on Research Assessment (DORA). (2012). San Francisco Declaration on Research Assessment. <https://sfedora.org/read/>
- Errington, T. M., et al. (2021). Investigating the replicability of preclinical cancer biology. *eLife*, 10, e71601. <https://doi.org/10.7554/eLife.71601>
- Fanelli, D. (2010). Do pressures to publish increase scientists' bias? An empirical support from US states data. *PLOS ONE*, 5(4), e10271.  
<https://doi.org/10.1371/journal.pone.0010271>
- Fanelli, D. (2012). Negative results are disappearing from most disciplines and countries. *Scientometrics*, 90, 891–904. <https://doi.org/10.1007/s11192-011-0494-7>
- Fortunato, S., et al. (2018). Science of science. *Science*, 359(6379), eaao0185.  
<https://doi.org/10.1126/science.aao0185>
- Halling, A., & Baekgaard, M. (2024). Administrative Burden in Citizen–State Interactions: A Systematic Literature Review. *Journal of Public Administration Research and Theory*, 34(2), 180–195. <https://doi.org/10.1093/jopart/muad023>
- Hicks, D., Wouters, P., Waltman, L., de Rijcke, S., & Rafols, I. (2015). Bibliometrics: The Leiden Manifesto for research metrics. *Nature*, 520, 429–431.  
<https://doi.org/10.1038/520429a>
- Ioannidis, J. P. A. (2005). Why most published research findings are false. *PLOS Medicine*, 2(8), e124. <https://doi.org/10.1371/journal.pmed.0020124>
- Kuhn, T. S. (1962). *The Structure of Scientific Revolutions*. University of Chicago Press.
- Lakatos, I. (1978). *The Methodology of Scientific Research Programmes*. Cambridge University Press.
- Latour, B., & Woolgar, S. (1979). *Laboratory Life: The Construction of Scientific Facts*. Sage.
- Longino, H. E. (1990). *Science as Social Knowledge*. Princeton University Press.
- Merton, R. K. (1973). *The Sociology of Science: Theoretical and Empirical Investigations*. University of Chicago Press.

- Munafò, M. R., et al. (2017). A manifesto for reproducible science. *Nature Human Behaviour*, 1, 0021. <https://doi.org/10.1038/s41562-016-0021>
- National Academies of Sciences, Engineering, and Medicine. (2019). *Reproducibility and Replicability in Science*. National Academies Press.  
<https://doi.org/10.17226/25303>
- National Science Foundation & Institute of Education Sciences. (2018). *Companion Guidelines on Replication & Reproducibility in Education Research: A Supplement to the Common Guidelines for Education Research and Development*.
- Nosek, B. A., et al. (2015). Promoting an open research culture. *Science*, 348(6242), 1422–1425. <https://doi.org/10.1126/science.aab2374>
- Open Science Collaboration. (2015). Estimating the reproducibility of psychological science. *Science*, 349(6251), aac4716. <https://doi.org/10.1126/science.aac4716>
- Popper, K. R. (1959). *The Logic of Scientific Discovery*. Hutchinson.
- Simmons, J. P., Nelson, L. D., & Simonsohn, U. (2011). False-positive psychology: Undisclosed flexibility in data collection and analysis allows presenting anything as significant. *Psychological Science*, 22(11), 1359–1366.  
<https://doi.org/10.1177/0956797611417632>
- Smaldino, P. E., & McElreath, R. (2016). The natural selection of bad science. *Royal Society Open Science*, 3, 160384. <https://doi.org/10.1098/rsos.160384>
- Soderberg, C. K., et al. (2021). Initial evidence of research quality of registered reports compared with the standard publishing model. *Nature Human Behaviour*, 5, 990–997. <https://doi.org/10.1038/s41562-021-01142-4>
- Whitley, R. (2000). *The Intellectual and Social Organization of the Sciences* (2nd ed.). Oxford University Press.
- Ziman, J. (2000). *Real Science: What It Is, and What It Means*. Cambridge University Press.